STAFF REVIEW · H446 226 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Initialises an object when it is created.

2. Left, root, right.

3. First In, First Out.

4. L then R.

I Do — source worked steps

1. Predict first — PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.

2. Model the reasoning — Return to10, traverse right child15 and print15, then print10. Output 2,5,15,10: the print follows both child calls. Node has Key, Left and Right fields.

We Do — guided CHECK

1. Model the reasoning
Visit 10 and enqueue child 15, then visit 30; final 15 gives 20,10,30,15.

You Do — source mark schemes

1. [3] Move the print after both recursive calls [1]; output 3,9,6 [1]; the parent is visited only after both child subtrees [1].

2. [5] Breadth-first 12,5,17,2,8 [2]; post-order 2,8,5,17,12 [3].

3. [6] If root is not null, enqueue(root) [1]; while queue nonempty [1], dequeue and visit [1], enqueue non-null left then right children [2]; queue maintains FIFO order across levels [1].

Exit Ticket — source answer

1. False. It uses a FIFO queue.